Building Blocks of an AC EV Charger

Introduction

As electric vehicles (EVs) become an increasingly common sight on our roads, understanding the intricate systems that power them is more crucial than ever. The charging infrastructure, in particular, forms the backbone of the EV ecosystem, enabling these vehicles to operate efficiently and reliably. This lecture delves into the fundamental components that constitute an EV charger, with a specific focus on the architecture of an AC EV charger. We begin by recalling that EVs utilize various battery configurations, ranging from low-voltage systems (48-72V) requiring 40-120V charging, to high-voltage systems (350-500V) that are prevalent in modern battery-powered electric vehicles. The charging process itself typically follows a Constant Current (CC) phase, followed by a Constant Voltage (CV) phase, sometimes incorporating multi-step constant current charging, especially for the lithium-ion batteries commonly found in EVs.

The necessity of a dedicated EV charger stems from several critical requirements. Firstly, it must precisely meet the specific charging profile dictated by the vehicle's Battery Management System (BMS), ensuring optimal battery health and longevity. Secondly, it provides essential safety protections, safeguarding both the vehicle and the user. Lastly, chargers often incorporate metering capabilities for billing and data collection. These dedicated systems are indispensable for the safe, efficient, and intelligent charging of electric vehicles. This lecture will break down the complex system into its core building blocks, starting with a general overview and then focusing on the specific design of AC EV chargers.

Core Concepts

Term	Definition	Significance
`Power Conversion Unit`	A system responsible for converting electrical energy from one form to another, typically from AC to DC for EV charging.	Essential for matching the grid's AC supply to the battery's DC requirements, ensuring efficient energy transfer.
`AC-DC Converter`	An electronic circuit that transforms alternating current (AC) into direct current (DC).	The initial stage of power conversion in an EV charger, rectifying grid power for battery compatibility.

Term	Definition	Significance
`Isolated DC-DC Converter`	A DC-DC converter that provides electrical isolation between its input and output circuits.	Crucial for safety, preventing electrical hazards, and allowing precise control over the charging voltage and current profile.
`EVSE (Electric Vehicle Supply Equipment)`	The physical equipment that supplies electrical energy for charging plug-in electric vehicles.	Acts as the interface between the grid and the vehicle, managing safety, communication, and power delivery.
`Onboard Charger (OBC)`	The power conversion unit located inside the electric vehicle, responsible for converting AC input to DC for the battery.	Enables AC charging by performing the necessary power conversion within the vehicle, making it a key component for AC charging.
`BMS (Battery Management System)`	An electronic system that manages a rechargeable battery, monitoring its state, calculating secondary data, and protecting it.	The "master" controller for battery charging, dictating the charging profile and ensuring battery safety and longevity.
`Communication Port`	A dedicated channel or interface for exchanging information between the charger and the EV.	Facilitates critical data exchange (e.g., charging status, battery health, safety protocols) for intelligent and safe charging.
`Connectors and Cables`	The physical interface (plug and cable) that connects the EVSE to the vehicle's inlet port.	Ensures physical and electrical compatibility between the charger and the vehicle, adhering to safety and power delivery standards.
	A backend system used by charging station service	Enables remote monitoring, billing, and overall operational

Term	Definition	Significance
`CMS (Central Management System)`	providers to monitor, manage, and control charging stations.	management of a network of EV charging stations.

Detailed Analysis

1. General Building Blocks of an EV Charger

At a high level, any EV charger, whether AC or DC, comprises several fundamental building blocks designed to facilitate the safe and efficient transfer of energy from the grid to the vehicle's battery. These blocks work in concert to manage power conversion, communication, and physical connection.

- **Power Conversion Unit:** This is the heart of any charger, primarily because the grid supplies Alternating Current (AC), while EV batteries require Direct Current (DC). Therefore, a power conversion unit is essential to transform AC into DC.
- It typically includes an **AC-DC converter** to rectify the incoming AC power.
- Following this, an **isolated DC-DC converter** is often employed. This component not only provides a controlled DC output but also ensures electrical isolation between the grid and the battery, which is a critical safety feature. The isolated DC-DC converter is crucial for precisely following the charging profiles determined by the BMS.
- Various **filters** are integrated within the power conversion unit. These filters ensure that the current delivered to the battery is ripple-free, which is vital for battery health. Additionally, filters are necessary to comply with Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC) requirements, preventing interference with other electronic devices.
- **Communication Ports:** For safe and intelligent charging, continuous communication between the charger and the EV is paramount. Dedicated communication channels or ports facilitate the exchange of vital information.
- These can include signals like **proximity pilot** and **control pilot**, which are used to establish a connection and manage the charging process.
- More advanced communication protocols, such as **CAN (Controller Area Network)**, with dedicated CAN high and CAN low pins, enable robust data exchange between the charger and the vehicle's internal systems. This

communication ensures that the charger understands the vehicle's status and the battery's needs at every instance.

- **EVSE (Electric Vehicle Supply Equipment):** The EVSE is the overarching equipment that sits between the grid and the vehicle. It can be an AC EVSE or a DC EVSE, depending on the type of charger.
- The EVSE integrates the power conversion unit (if it's a DC charger) and the communication interfaces.
- It also houses safety mechanisms and metering functionalities.
- **Connectors and Cables:** These are the physical links that connect the EVSE to the vehicle.
- A **charging plug** extends from the EVSE via a cable.
- This plug connects to the **vehicle inlet connector** on the EV.
- Crucially, the form factors (size, shape) of the charging plug and the vehicle inlet connector must match to ensure compatibility across different vehicle manufacturers.
- The **cables** themselves must be robust and of appropriate dimensions, including length (commonly 5 meters, 7.5 meters, or 10 meters), to safely carry the required power. The design and selection of these components are critical for user convenience and safety.

2. Architecture of an AC EV Charger

When we specifically consider an AC EV charger, the distribution of these building blocks changes significantly compared to a DC charger. The defining characteristic of an AC charger is that the primary power conversion from AC to DC occurs inside the vehicle, not within the external charging station.

- **The AC EVSE (AC Electric Vehicle Supply Equipment):**
- The AC EVSE receives input directly from the electrical grid, which can be either single-phase or three-phase, depending on the power level of the charger. AC chargers typically range from 2 kW to 22 kW.
- Crucially, the output of the AC EVSE is also AC. This AC power is directly supplied to the vehicle.
- Inside the AC EVSE, several components ensure safe and monitored operation:
- **Safety Interlocks/Devices:** These include contactors and relays that can quickly disconnect the power supply in case of a fault or if the vehicle's controller requests a shutdown.

- **Controller:** A dedicated controller (sometimes called a supply equipment communication controller or charger controller) manages the overall operation of the AC EVSE, including communication with the vehicle and the central management system.
- **Sensors:** Voltage and current sensors continuously monitor the input power from the grid and the AC power being delivered to the vehicle. This data is used for metering purposes, protection, and to inform the controller.
- It is vital to understand that the AC EVSE does not perform any power conversion. Its primary functions are metering, monitoring the vehicle's status, and ensuring safety by controlling the power flow based on communication with the vehicle.
- **Components Inside the Vehicle (Onboard Charger - OBC):**
- Once the AC power enters the vehicle through the inlet connector, it first goes to a **rectifier**. This rectifier converts the incoming AC into DC.
- Following the rectifier, an **isolated DC-DC converter** takes over. This converter is responsible for providing a controlled DC output, which is essential for following the specific charging profile (CC-CV) required by the battery. The isolation feature also ensures safety.
- **Protective Devices:** Similar to the EVSE, the vehicle also has its own protective devices (contactors, relays) to safeguard the battery and vehicle electronics.
- The final DC output from the isolated DC-DC converter, after passing through protective devices, is then fed to the **EV battery** for charging. The operation of the isolated DC-DC converter is precisely controlled to achieve the desired CC or CV charging.
- **The Role of the Battery Management System (BMS):**
- The **BMS** is the "master" controller within the vehicle. It continuously monitors critical parameters of the battery cells, such as voltage, current, and temperature.
- Based on these measurements and the battery's state of charge, the BMS determines the optimal charging profile.
- The BMS communicates its requirements (e.g., desired voltage and current at the output of the DC-DC converter) to the controller of the vehicle's power conversion unit (the onboard charger). This controller then adjusts the duty ratios or phase shifts of the power electronic switches in the rectifier and DC-DC converter to achieve the desired output.

- Furthermore, the BMS communicates with the AC EVSE controller via a dedicated communication channel (e.g., control pilot pin in the connector). This communication informs the AC EVSE whether the vehicle is ready to accept a charge, if the battery is full, or if any safety issues arise. The BMS essentially tells the AC EVSE when to supply power and when to cut it off.
- **Communication Ecosystem:**
- **BMS ↔ Onboard Charger Controller:** Internal communication within the vehicle to manage the power conversion process according to battery needs.
- **BMS ↔ AC EVSE Controller:** External communication to coordinate the charging session, including readiness, safety status, and termination signals.
- **AC EVSE Controller ↔ CMS (Central Management System):** The AC EVSE also communicates with a central management system, typically operated by the charging station service provider. This communication is crucial for metering, monitoring the charging station's status, and overall operational management.

In summary, for an AC EV charger, the AC EVSE acts primarily as a smart switch and monitoring unit, delivering AC power to the vehicle. The actual power conversion from AC to DC, which is essential for charging the battery, takes place within the vehicle itself, handled by the onboard charger under the precise command of the BMS.

Key Takeaways

- [ ] EV chargers are essential for converting grid AC power to battery-compatible DC power, meeting BMS requirements, and ensuring safety.
- [ ] General EV charger building blocks include a Power Conversion Unit (AC-DC, isolated DC-DC, filters), Communication Ports, EVSE, and Connectors/Cables.
- [ ] AC EVSEs supply AC power directly to the vehicle and do NOT perform AC-DC power conversion.
- [ ] The AC EVSE's primary functions are metering, monitoring vehicle status, and ensuring safety through interlocks and communication.
- [ ] In an AC charging scenario, the actual AC-DC power conversion occurs inside the vehicle, within the Onboard Charger (OBC).
- [ ] The Onboard Charger consists of a rectifier and an isolated DC-DC converter, controlled to follow the battery's charging profile.

- [ ] The Battery Management System (BMS) acts as the "master" controller, monitoring battery health and dictating charging parameters to both the OBC and the AC EVSE.
- [ ] Robust communication channels between the BMS, OBC controller, AC EVSE controller, and CMS are critical for safe and efficient charging.

Conclusion

The architecture of an AC EV charger reveals a sophisticated interplay of power electronics, control systems, and communication protocols. While the external AC EVSE provides the necessary interface to the grid, handling safety, metering, and initial communication, the true power conversion intelligence resides within the vehicle's onboard charger, orchestrated by the Battery Management System. This distributed approach to power conversion is a defining characteristic of AC charging, allowing for a simpler, often more cost-effective external charging station.

Understanding these building blocks is fundamental for anyone involved in the design, deployment, or maintenance of EV charging infrastructure. The seamless integration of these components ensures that electric vehicles can be charged safely, efficiently, and in a manner that preserves battery longevity. As the EV landscape continues to evolve, the principles governing these core components will remain central to developing even more advanced and user-friendly charging solutions.